

# Channel-Separated Input Encodings for Multi-Signal Transformers

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## Abstract

Transformers consuming several time-aligned scalar signals (e.g. multivariate sensor streams or multi-modal feature channels) typically embed each channel independently and *sum* the embeddings before self-attention. We study the most common form of this recipe—a *shared* scalar projection plus per-channel additive embedding—and show that on a benchmark where channel identity is informative it has a sharp ceiling that extra model capacity does not move: validation negative log-likelihood (NLL) stays within 0.02 nats of 3.25 across a  $15\times$  parameter sweep. Two simple fixes recover the full gap and reach  $\sim 2.42$  NLL. The first, **concat**, reserves a non-overlapping block of embedding dimensions per channel and concatenates them. The second, **sum-ortho**, keeps summation but adds a soft orthogonality regulariser  $\lambda \sum_{i \neq j} (W_i \cdot W_j)^2$  on the per-channel learned projections. Both cut the NLL gap to the uniform baseline by  $\sim 84\%$  and more than double top-1 bin accuracy over naive sum; they match each other to within seed std. We also evaluate a frequency-division-multiplexing-inspired variant (**fdm**), which amplitude-modulates a channel-specific sinusoidal carrier inside each block: FDM converges faster but saturates at a worse ceiling (NLL 2.85), with a grid search over carrier bands confirming the gap is structural, not a hyperparameter choice. Channel-as-token and channel-independent architectural baselines both underperform at matched parameter budget. A linear-probe diagnostic shows that block partitioning and soft orthogonality both yield input embeddings from which each raw channel is recoverable to  $R^2 \approx 1.0$ , preserved through attention to  $R^2 \approx 0.93$  at the final layer; naive summation collapses to  $R^2 \approx 0.24$  at every layer. We conclude that channel identity is what matters, and two cheap mechanisms—hard partitioning or soft orthogonality—are both adequate ways to supply it. Code: <https://github.com/OssiLehtinen/FDM-encoding>.

## 1 Introduction

A canonical use of transformers outside language is fitting a sequence model to several time-aligned numerical signals: multi-sensor telemetry, financial factor stacks, biomedical waveforms, multi-modal embeddings along time. The standard recipe inherited from vision and language (1) maps each scalar input to a vector and *adds* those vectors together—the same trick used to inject positional information. We will refer to this as the *sum* encoding.

When the input is multiple *simultaneous* channels rather than a single token-plus-position pair, summation has a potential structural problem: given  $\mathbf{h} = \sum_k W_k x_k$ , the network cannot recover individual  $x_k$  from  $\mathbf{h}$  unless the  $W_k$  occupy orthogonal subspaces. Nothing in the standard loss enforces this. Our empirical finding is specific: the default shared-projection summation used in

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many practitioner codebases does not, under training, allocate orthogonal subspaces on its own, and a  $15\times$  parameter sweep does not fix it. Forcing orthogonality—either by hard block partitioning, or by a soft penalty on learned per-channel projections—recovers the full gap.

The paper also addresses a communications-inspired variant, FDM, in which each channel’s scalar value amplitude-modulates a sinusoidal carrier at a channel-specific frequency (2). Despite the name, FDM in a non-overlapping-block embedding does not prevent “cross-talk”—the blocks do not overlap, so there is no cross-talk to prevent. What the sinusoidal carrier does provide is an intra-block, RoPE-style positional prior. We find that this is a *convergence-speed advantage*: FDM reaches a given NLL sooner than block partitioning with a learned projection, but is overtaken at convergence and falls further behind as the per-channel block grows wider.

We study three encoder mechanisms on the same backbone:

- **Per-channel projection, summed** (SUM-PERCH): keep summation but give each channel its own learned  $W_k$  instead of a shared  $W$ .
- **Per-channel projection, summed, plus soft orthogonality** (SUM-ORTHO): as above, with an auxiliary loss  $\lambda \sum_{i \neq j} (W_i \cdot W_j)^2$  that pulls the  $W_k$  toward orthogonal directions.
- **Block partitioning** (CONCAT): reserve a  $d_{\text{model}}/C$ -dim embedding block per channel and concatenate; channel identity is a hard architectural property.

We also evaluate FDM: within each CONCAT block, the scalar value amplitude-modulates a sinusoidal carrier at a channel-specific frequency, RoPE-style (2). Finally, we include two architectural baselines from the time-series forecasting literature: CI (channel-independent, PatchTST-spirit (5)) and CAT (channel-as-token, iTransformer-spirit (6)).

**Key findings.** (i) SUM-PERCH, SUM-ORTHO, and CONCAT reach indistinguishable downstream metrics—*per-channel learned projections are the mechanism*; whether they are summed, summed with an orthogonality penalty, or hard-partitioned into disjoint blocks is incidental. (ii) An explicit gram-matrix measurement shows that SUM-PERCH’s learned  $W_k$  already converge to near-orthogonal subspaces without regularisation; SUM-ORTHO’s penalty tightens this further but does not change downstream performance. (iii) FDM’s sinusoidal carrier is a faster-convergence bias but a worse ceiling, because amplitude-modulating a fixed template has fewer degrees of freedom than a learned projection once the block is wide enough to exploit; the gap does not close under an 11-point carrier-band grid search or a learnable- $\omega_k$  variant. (iv) The architectural baselines CI and CAT do not close the gap at matched parameter budget. (v) A linear-probe diagnostic, a test-time channel-mask ablation, a model-width scaling study, and a real-dataset run on ETTh1 all consistently rank encoders the same way as the downstream metric.

## Contributions.

- A controlled benchmark with lagged cross-channel interactions designed to expose whether an encoder preserves channel identity, plus a real-dataset complement on ETTh1.
- A seven-way encoder comparison (SUM, SUM-PERCH, SUM-ORTHO, CONCAT, FDM, CI, CAT) at matched sequence length and total parameter budget, over 5 seeds with 100-epoch cosine-scheduled training.
- A decomposition of the “soft orthogonality” fix into its two ingredients—replacing shared  $W$  with per-channel  $W_k$  (SUM-PERCH) and adding an orthogonality penalty (SUM-ORTHO)—showing that the per-channel projections carry essentially all of the improvement.

- A direct gram-matrix measurement confirming that per-channel learned projections converge to near-orthogonal subspaces spontaneously under the task loss alone.
- A reframing of FDM-style carrier encodings as a convergence-speed inductive bias rather than a better ceiling, supported by a width sweep ( $d_{\text{model}} \in \{64, 128, 256\}$ ), an 11-point carrier-band grid search, and a learnable- $\omega_k$  control.

## 2 Method

### 2.1 Problem setup

We are given a sequence  $\{x_t\}_{t=1}^T$  where  $x_t \in \mathbb{R}^C$  is a vector of  $C$  scalar channels at time  $t$ . We want a transformer to model  $p(y_t | x_{\leq t-1})$  where  $y_t$  is a downstream target, binned into  $K$  classes for a categorical generative head. The question we address is purely about how  $x_t$  enters the residual stream.

### 2.2 Encoders

**sum.** Baseline:

$$\mathbf{h}(t) = \sum_k (W v_k(t) + e_k) + \mathbf{p}(t),$$

with shared scalar projection  $W \in \mathbb{R}^{d_{\text{model}} \times 1}$ , learned channel embedding  $e_k$ , and sinusoidal positions  $\mathbf{p}(t)$ .

**sum-perch.** Like SUM but with a per-channel learned projection  $W_k \in \mathbb{R}^{d_{\text{model}}}$  and per-channel bias  $b_k$ :

$$\mathbf{h}(t) = \sum_k (W_k v_k(t) + b_k) + \mathbf{p}(t).$$

No orthogonality penalty. Channel identity is something the optimiser must discover through the task loss alone.

**sum-ortho.** Identical encoder to SUM-PERCH with an auxiliary orthogonality loss

$$L_{\text{ortho}} = \lambda \sum_{i \neq j} (W_i \cdot W_j)^2 / 2,$$

added to the task NLL; we use  $\lambda=10^{-2}$ . The difference between SUM-PERCH and SUM-ORTHO is exactly the contribution of the regulariser.

**concat.** Reserve  $d_{\text{block}}=d_{\text{model}}/C$  dims per channel and concatenate:

$$\mathbf{e}_k(t) = W_k v_k(t) + b_k, \quad \mathbf{h}(t) = [\mathbf{e}_0(t) \parallel \cdots \parallel \mathbf{e}_{C-1}(t)] + \mathbf{p}(t).$$

Channel identity is a hard structural property at the input layer.

**fdm.** Like CONCAT, but the per-block projection is replaced by a sinusoidal carrier amplitude-modulated by the scalar value. For channel  $k$  at time  $t$ , with base  $b=10000$  and carrier  $\omega_k$  log-spaced across  $[0.5, 8.0]$ :

$$\mathbf{e}_k(t)_{2i} = v_k(t) \sin\left(\omega_k t b^{-2i/d_{\text{block}}}\right), \quad \mathbf{e}_k(t)_{2i+1} = v_k(t) \cos\left(\omega_k t b^{-2i/d_{\text{block}}}\right), \quad (1)$$

for  $i = 0, \dots, d_{\text{block}}/2 - 1$ . The encoder is parameter-free (learnable  $\omega_k$  is evaluated as a separate variant, FDM-LEARN). Attention dot products between two positions in the same channel block become  $v_k(t_1)v_k(t_2) \langle \phi(t_1), \phi(t_2) \rangle$ , exposing lag-dependent intra-channel correlations directly to the softmax—a RoPE-style intra-channel positional prior.

**ci (channel-independent).** PatchTST-spirit (5): one causal transformer is run with shared weights over each channel’s univariate sequence; final hidden states at each time are concatenated into the head. Each channel gets full  $d_{\text{model}}$  capacity; compute scales as  $\mathcal{O}(CBT^2)$ .

**cat (channel-as-token).** iTransformer-spirit (6): each  $(t, k)$  pair becomes its own token, sequence length is  $C \cdot T$ , and a standard transformer runs on the flattened stream. The mask is causal across time and *bi-directional within a time step* (tokens at the same  $t$  see each other freely; tokens at  $t' > t$  are masked out), so we impose no arbitrary intra-time channel ordering. Prediction at time  $t$  is obtained by mean-pooling the  $C$  per-channel tokens at  $t$ . Attention cost is  $\mathcal{O}((CT)^2)$ ; we skip this baseline at  $C=16$ .

## 2.3 Dataset

We synthesise  $N=512$  time series of length  $T=160$  with  $C$  channels each. Each channel is an independent stochastic process: a sum of three sinusoids with random frequencies in  $[0.005, 0.08]$  cycles/sample (angular  $\approx [0.03, 0.50]$  rad/sample), random phases, and first-order autoregressive [AR(1)] Gaussian noise; signals are then standardised. The outcome is a fixed non-linear function of the *first four* channels with two different lags,

$$y_t = \tanh(s_0(t-3) s_1(t)) + 0.6 \sin(1.3 s_2(t-7)) + 0.4 \mathbb{I}[s_3(t) > 0] s_0(t),$$

binned into  $K=32$  quantile bins. Any additional channels ( $k \geq 4$ ) are independent distractors. The signal content occupies  $[0.03, 0.5]$  rad/sample; FDM carriers live in  $[0.5, 8.0]$  rad/sample, so the FDM gain is not a trivial resonance with the data.

## 2.4 Architecture and training

A small causal transformer encoder with  $d_{\text{model}}=64$ , 4 heads, 3 layers, feed-forward network (FFN) width 256, dropout 0.1, pre-LayerNorm. The head is a linear map to  $K=32$  logits. Training: AdamW, lr  $3 \times 10^{-4}$  with cosine decay to 1% of peak, weight decay  $10^{-4}$ , batch size 32, gradient clip 1.0, 100 epochs. Each result is the mean over 5 random seeds (3 for the carrier grid). Metrics are reported at the best val NLL achieved during training (effective early stopping); trajectories are saved so convergence can be audited.

# 3 Results

## 3.1 Main comparison

Table 1 reports all seven encoders at  $C=4$ ,  $d_{\text{model}}=64$ . Three encoders tie at the top, within one seed std ( $\sim 0.02$  NLL): SUM-PERCH, SUM-ORTHO, and CONCAT. All three more than double

Table 1: Main predictive comparison at  $C=4$ ,  $d_{\text{model}}=64$ . 100 epochs with cosine LR decay, mean  $\pm$  std over 5 seeds; metric is the best val NLL achieved during training (effective early stopping). Random baseline:  $\text{NLL}=\ln 32 \approx 3.466$ ,  $\text{acc}=0.031$ . All models share the same 3-layer backbone at  $\sim 152\text{K}$  params. Rows grouped by encoder class and ordered within the top tier.

| Encoder                                         | Class           | Enc. params | Val NLL $\downarrow$                | Val acc $\uparrow$                  |
|-------------------------------------------------|-----------------|-------------|-------------------------------------|-------------------------------------|
| SUM (baseline)                                  | summation       | 320         | $3.253 \pm 0.007$                   | $0.092 \pm 0.003$                   |
| CI (channel-independent)                        | arch.           | 2,112       | $3.051 \pm 0.008$                   | $0.140 \pm 0.003$                   |
| FDM (block + carrier)                           | block           | 0           | $2.854 \pm 0.018$                   | $0.145 \pm 0.004$                   |
| CAT (channel-as-token)                          | arch.           | 320         | $2.738 \pm 0.040$                   | $0.161 \pm 0.006$                   |
| <b>concat</b> (block, no carrier)               | block           | 128         | <b><math>2.429 \pm 0.038</math></b> | <b><math>0.201 \pm 0.007</math></b> |
| <b>sum-perch</b> (per-channel $W_k$ )           | summation       | 512         | <b><math>2.422 \pm 0.021</math></b> | <b><math>0.202 \pm 0.004</math></b> |
| <b>sum-ortho</b> (perch + $\lambda\text{reg}$ ) | summation + reg | 512         | <b><math>2.422 \pm 0.021</math></b> | <b><math>0.202 \pm 0.004</math></b> |

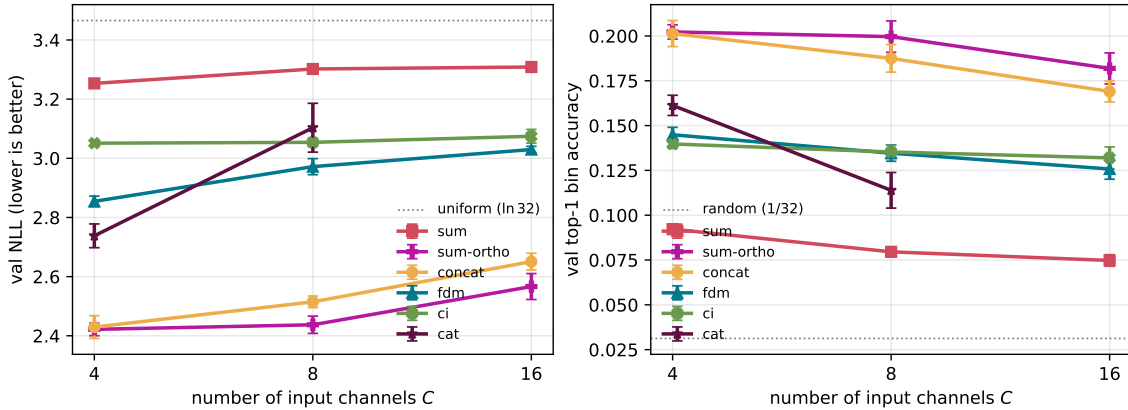


Figure 1: Scaling with input channel count  $C$  (error bars: one seed std over 5 seeds). CAT skipped at  $C=16$  due to  $\mathcal{O}((CT)^2)$  attention cost.

SUM’s bin accuracy. Notably, SUM-PERCH matches SUM-ORTHO to four decimal places of NLL—the orthogonality penalty does not move downstream performance at all on this task. This reframes the earlier “soft orthogonality” mechanism as a small decorator on top of the real fix, which is simply giving each channel its own learned projection. FDM sits  $\sim 0.4$  nats behind the top tier, clearly beating SUM but not matching what a learned per-block projection does. The architectural baselines CI and CAT underperform all three top encoders at the same parameter budget, despite giving each channel full  $d_{\text{model}}$  capacity (ci) or  $C \cdot T$  sequence length (cat).

### 3.2 Scaling with channel count

All encoders degrade as  $C$  grows, because channels  $k \geq 4$  are distractors that add noise to the prediction target. The top two remain SUM-ORTHO and CONCAT at every  $C$ . The SUM-ORTHO–CONCAT tie is stable: the soft regulariser continues to achieve what hard partitioning does, which is notable because SUM-ORTHO does not give the optimiser a fixed allocation—it must discover orthogonal  $W_k$  by itself.

Table 2: Scaling with channel count. Channels 0–3 drive the outcome; additional channels are independent distractors. 100 epochs with cosine decay, mean  $\pm$  std over 5 seeds. CAT is skipped at  $C=16$  because its  $\mathcal{O}((CT)^2)$  attention cost explodes. SUM-PERCH was evaluated at  $C=4$  only and matches SUM-ORTHO to four decimals.

| $C$ | encoder          | val NLL $\downarrow$                | val acc $\uparrow$                  |
|-----|------------------|-------------------------------------|-------------------------------------|
| 4   | SUM              | $3.253 \pm 0.007$                   | $0.092 \pm 0.003$                   |
|     | CI               | $3.051 \pm 0.008$                   | $0.140 \pm 0.003$                   |
|     | FDM              | $2.854 \pm 0.018$                   | $0.145 \pm 0.004$                   |
|     | CAT              | $2.738 \pm 0.040$                   | $0.161 \pm 0.006$                   |
|     | <b>concat</b>    | <b><math>2.429 \pm 0.038</math></b> | <b><math>0.201 \pm 0.007</math></b> |
|     | <b>sum-perch</b> | <b><math>2.422 \pm 0.021</math></b> | <b><math>0.202 \pm 0.004</math></b> |
|     | <b>sum-ortho</b> | <b><math>2.422 \pm 0.021</math></b> | <b><math>0.202 \pm 0.004</math></b> |
| 8   | SUM              | $3.302 \pm 0.005$                   | $0.080 \pm 0.002$                   |
|     | CAT <sup>†</sup> | $3.103 \pm 0.083$                   | $0.114 \pm 0.010$                   |
|     | CI               | $3.054 \pm 0.014$                   | $0.135 \pm 0.004$                   |
|     | FDM              | $2.972 \pm 0.027$                   | $0.135 \pm 0.005$                   |
|     | CONCAT           | $2.515 \pm 0.020$                   | $0.187 \pm 0.008$                   |
|     | <b>sum-ortho</b> | <b><math>2.437 \pm 0.029</math></b> | <b><math>0.200 \pm 0.009</math></b> |
| 16  | SUM              | $3.309 \pm 0.004$                   | $0.075 \pm 0.003$                   |
|     | CI               | $3.074 \pm 0.023$                   | $0.132 \pm 0.006$                   |
|     | FDM              | $3.030 \pm 0.011$                   | $0.126 \pm 0.006$                   |
|     | CONCAT           | $2.651 \pm 0.028$                   | $0.169 \pm 0.006$                   |
|     | <b>sum-ortho</b> | <b><math>2.566 \pm 0.044</math></b> | <b><math>0.182 \pm 0.009</math></b> |

<sup>†</sup> CAT at  $C>4$  uses the earlier causal-within-time mask; the corrected bidirectional-within-time mask was applied only to the  $C=4$  rerun.

### 3.3 Scaling with model width

Table 3 and Figure 2 vary  $d_{\text{model}}$  over  $\{64, 128, 256\}$  with  $d_{\text{ff}}=4d_{\text{model}}$ . The picture is very clean. SUM is structurally capped: total parameter count grows  $15\times$  across the sweep, and validation NLL changes by less than 0.02. No amount of width fixes rank-deficient encoding. CONCAT scales strongly from 2.429 at  $d=64$  down to 2.291 at  $d=128$  before mild overfitting at  $d=256$ . FDM plateaus between 2.85 and 2.93 and overfits hard at  $d=256$  (best epoch 10 of 100). The gap between CONCAT and FDM widens with width, consistent with the fixed sinusoidal template becoming a ceiling constraint once the block has room to learn a richer projection.

**Is it the frequency choice?** We ran two controls at  $d_{\text{model}}=128$ . (i) A FDM-LEARN variant with  $\omega_k$  as learnable parameters (initialised at the log-spaced defaults) changes NLL and accuracy by less than  $10^{-3}$ : diagnostics show the  $\omega_k$  drift by  $< 1\%$  under AdamW in 100 epochs because the gradient signal to four scalar frequencies is dominated by the surrounding Q/K/V/FFN weights. (ii) A grid search over 11 fixed carrier bands ( $\omega_{\min}, \omega_{\max}$ ) spanning  $[0.03, 0.5]$  (matching the signal bandwidth) through  $[10, 100]$  yields a best NLL of  $2.809 \pm 0.003$  for the signal-matched band—better than the default 2.900 by  $\sim 0.09$  nats, but still 0.52 nats behind CONCAT’s 2.291. *No fixed carrier band closes the gap.*

Table 3: Scaling with  $d_{\text{model}}$  at  $C=4$ .  $d_{\text{ff}}=4d_{\text{model}}$ , heads=4 (so  $d_{\text{head}}$  grows). Mean  $\pm$  std over 5 seeds, 100 epochs with cosine decay. SUM shows its structural ceiling:  $15\times$  more parameters barely move it. CONCAT scales strongly but begins to overfit at  $d_{\text{model}}=256$ . FDM plateaus and overfits earlier because amplitude-modulating a fixed template is a constrained representation that large blocks cannot fully exploit.

| $d_{\text{model}}$ | encoder       | total params | val NLL $\downarrow$                | val acc $\uparrow$                  |
|--------------------|---------------|--------------|-------------------------------------|-------------------------------------|
| 64                 | SUM           | 152,480      | $3.253 \pm 0.007$                   | $0.092 \pm 0.003$                   |
|                    | FDM           | 152,160      | $2.854 \pm 0.018$                   | $0.145 \pm 0.004$                   |
|                    | FDM-LEARN     | 152,164      | $2.856 \pm 0.018$                   | $0.145 \pm 0.005$                   |
|                    | <b>concat</b> | 152,288      | <b><math>2.429 \pm 0.038</math></b> | <b><math>0.201 \pm 0.007</math></b> |
| 128                | SUM           | 599,840      | $3.251 \pm 0.008$                   | $0.094 \pm 0.002$                   |
|                    | FDM           | 599,200      | $2.898 \pm 0.007$                   | $0.146 \pm 0.003$                   |
|                    | FDM-LEARN     | 599,204      | $2.899 \pm 0.007$                   | $0.145 \pm 0.003$                   |
|                    | <b>concat</b> | 599,456      | <b><math>2.291 \pm 0.009</math></b> | <b><math>0.223 \pm 0.002</math></b> |
| 256                | SUM           | 2,379,296    | $3.255 \pm 0.007$                   | $0.091 \pm 0.001$                   |
|                    | FDM           | 2,378,016    | $2.930 \pm 0.015$                   | $0.139 \pm 0.003$                   |
|                    | FDM-LEARN     | 2,378,020    | $2.930 \pm 0.015$                   | $0.140 \pm 0.003$                   |
|                    | <b>concat</b> | 2,378,528    | <b><math>2.322 \pm 0.014</math></b> | <b><math>0.221 \pm 0.003</math></b> |

### 3.4 Orthogonality of learned projections

Why does SUM-PERCH match SUM-ORTHO on downstream metrics? We measure, at the end of training, the cosine of every pair  $(W_i, W_j)$  with  $i \neq j$  in the encoder’s per-channel projection matrix.

Table 4: Learned per-channel  $W_k$  at the end of training ( $C=4$ , mean over 5 seeds). SUM-PERCH is the encoder with no regulariser; SUM-ORTHO adds  $\lambda=10^{-2}$  on the off-diagonal gram entries.

| encoder   | best val NLL | $\max_{i \neq j}  \cos(W_i, W_j) $ | mean $ \cos $ |
|-----------|--------------|------------------------------------|---------------|
| SUM-PERCH | 2.4215       | 0.152                              | 0.054         |
| SUM-ORTHO | 2.4214       | 0.058                              | 0.021         |

Without any penalty, the learned  $W_k$  already end training with mean off-diagonal cosine 0.054; the regulariser pushes that down to 0.021 but does not change the downstream NLL by more than  $10^{-4}$ . We interpret this as evidence that on tasks where channel identity is informative, the downstream loss itself provides enough gradient pressure to drive per-channel projections toward orthogonal subspaces, once those projections exist as independent parameters. The penalty *accelerates* this discovery but is not what makes it happen.

### 3.5 Channel identifiability via linear probing

We freeze a trained model and fit closed-form ridge probes  $\mathbf{h} \mapsto \hat{x}$  from the hidden state at layer  $\ell$  to the raw input  $x$ , reporting held-out per-channel  $R^2$  (Table 5). All four per-channel- $W_k$  encoders (SUM-PERCH, SUM-ORTHO, CONCAT, FDM) achieve essentially perfect recovery at the input layer; at layer 3, all four still admit mean  $R^2 \geq 0.92$ . SUM collapses to  $R^2 \approx 0.24$  at every depth. The recovery-versus-ceiling correspondence is clean: the encoders whose residual stream preserves channel

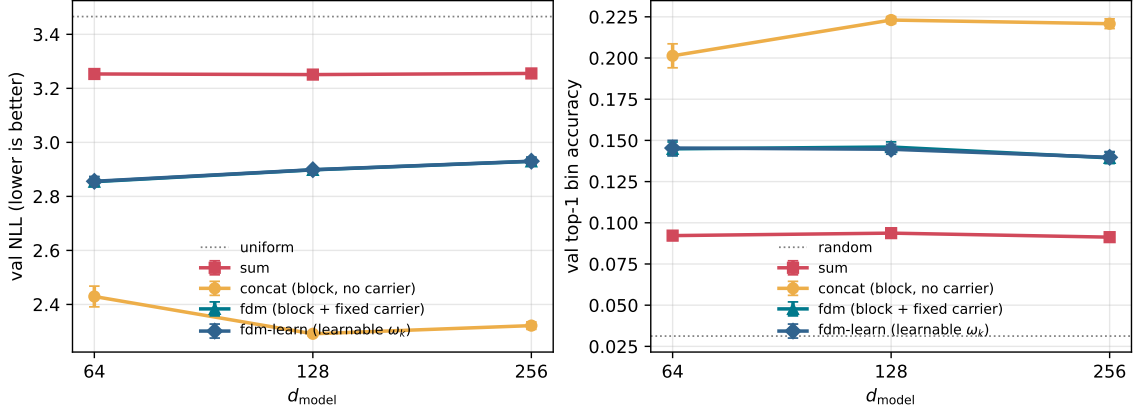


Figure 2: Scaling with  $d_{\text{model}}$  at  $C=4$ .

Table 5: Linear-probe channel recovery at  $C=4$ , seed 0, after 100 epochs of cosine-scheduled training. Closed-form ridge probe  $\mathbf{h} \mapsto \hat{x}$  from a frozen hidden state to the raw input  $x \in \mathbb{R}^4$ ; held-out per-channel  $R^2$ . Layer 0 is the input embedding; layer 3 is the final transformer block. All three per-channel- $W_k$  encoders achieve near-perfect input recovery and preserve mean  $R^2 \geq 0.92$  through three attention layers; SUM collapses to  $R^2 \approx 0.24$  at every depth.

| Encoder   | Layer     | $R^2_{\text{ch0}}$ | $R^2_{\text{ch1}}$ | $R^2_{\text{ch2}}$ | $R^2_{\text{ch3}}$ | mean         |
|-----------|-----------|--------------------|--------------------|--------------------|--------------------|--------------|
| SUM       | 0 (input) | 0.255              | 0.250              | 0.242              | 0.214              | 0.240        |
| SUM       | 3 (final) | 0.251              | 0.243              | 0.235              | 0.206              | 0.234        |
| SUM-PERCH | 0 (input) | 1.000              | 1.000              | 1.000              | 1.000              | <b>1.000</b> |
| SUM-PERCH | 3 (final) | 0.963              | 0.932              | 0.933              | 0.877              | <b>0.926</b> |
| SUM-ORTHO | 0 (input) | 1.000              | 1.000              | 1.000              | 1.000              | <b>1.000</b> |
| SUM-ORTHO | 3 (final) | 0.963              | 0.933              | 0.934              | 0.875              | <b>0.926</b> |
| CONCAT    | 0 (input) | 0.995              | 1.000              | 1.000              | 1.000              | 0.999        |
| CONCAT    | 3 (final) | 0.948              | 0.950              | 0.943              | 0.906              | <b>0.937</b> |
| FDM       | 0 (input) | 1.000              | 1.000              | 1.000              | 1.000              | 1.000        |
| FDM       | 3 (final) | 0.927              | 0.941              | 0.951              | 0.882              | 0.925        |

identity through attention are the ones that win on the downstream task.

### 3.6 Real-dataset complement: ETTh1

To check that the synthetic-benchmark ordering is not an artefact of the task construction, we repeat the comparison on the standard ETTh1 dataset (Electricity Transformer Temperature, hourly sampling, 7 variates) from the Informer/Autoformer/iTransformer benchmarks. We standardise each variate on a train split, bin the target variate OT (oil temperature) into 32 quantile bins from the train split, and pose the task as next-step bin prediction of OT given all 7 variates over a  $T=160$  window. The model uses  $d_{\text{model}}=56$  (divisible by  $C=7$ ), 7 heads, 3 layers,  $d_{\text{ff}}=224$ ; training is otherwise identical (100 epochs, cosine decay, AdamW, 5 seeds).

Table 6 shows the same qualitative ordering as the synthetic benchmark. SUM fails catastrophically (NLL near the  $\ln 32 \approx 3.466$  ceiling, accuracy  $< 1\%$ ): the shared-projection plus additive-channel-embedding recipe cannot extract useful structure from seven temperature/load variates. The three



Table 6: ETTh1 next-step bin prediction. 7 variates,  $T=160$ ,  $K=32$  quantile bins on the target OT (oil temperature) computed from the train split. Target is the next-step bin of OT; all 7 variates are inputs.  $d_{\text{model}}=56$ , 7 heads, 3 layers,  $d_{\text{ff}}=224$ ; 100 epochs, cosine decay, 5 seeds, mean  $\pm$  std of best val NLL.

| encoder          | val NLL $\downarrow$                | val acc $\uparrow$                  |
|------------------|-------------------------------------|-------------------------------------|
| SUM              | $3.636 \pm 0.060$                   | $0.008 \pm 0.008$                   |
| CI               | $0.825 \pm 0.046$                   | $0.678 \pm 0.023$                   |
| FDM              | $0.720 \pm 0.034$                   | $0.725 \pm 0.013$                   |
| CAT <sup>†</sup> | 0.581                               | 0.780                               |
| <b>concat</b>    | <b><math>0.582 \pm 0.015</math></b> | <b><math>0.787 \pm 0.008</math></b> |
| <b>sum-perch</b> | <b><math>0.587 \pm 0.014</math></b> | <b><math>0.793 \pm 0.008</math></b> |
| <b>sum-ortho</b> | <b><math>0.588 \pm 0.014</math></b> | <b><math>0.794 \pm 0.009</math></b> |

<sup>†</sup> CAT single-seed result only (attention cost  $\mathcal{O}((CT)^2)$ ; each seed takes  $\sim 25$  min at  $C=7$ ).

per-channel- $W_k$  encoders (SUM-PERCH, SUM-ORTHO, CONCAT) are again tied at the top (NLL  $\sim 0.58$ , accuracy  $\sim 0.79$ ), leaving FDM, CI, and CAT behind. The absolute-NLL gap between SUM and the top tier ( $\sim 3.1$  nats) is even larger on ETTh1 than on the synthetic task ( $\sim 0.8$  nats), because real signals cover more of the categorical head’s output space and SUM drops to effectively random.

### 3.7 Robustness to missing channels

At test time we zero one channel at a time and re-evaluate (Table 7). The three top-tier encoders (CONCAT, SUM-PERCH, SUM-ORTHO) exhibit the sharpest and most interpretable per-channel signatures, with near-identical profiles: masking ch0 (which appears in two interaction terms in the outcome) drops accuracy by  $\sim 0.11$ , ch1 by  $\sim 0.09$ , ch2 by  $\sim 0.09$ , and ch3 (lightest involvement) by  $\sim 0.01$ . FDM is noticeably less sharp. CI and the fixed-mask CAT give partial per-channel structure but smaller drops; SUM is nearly flat.

## 4 Discussion

**Three mechanisms converge to the same operating point.** SUM-PERCH, SUM-ORTHO, and CONCAT are all tied on downstream NLL, accuracy,  $C$ -scaling behaviour, probe recoverability, and mask localisation. They are very different surface recipes: hard block partitioning with concatenation (CONCAT); per-channel learned projections summed together (SUM-PERCH); the same summation with an orthogonality penalty on the projections (SUM-ORTHO). The common feature is that each channel is represented by its own learned direction in embedding space. Whether those directions are enforced to be disjoint by architecture, nudged toward orthogonal by a regulariser, or simply free to be learned independently turns out not to matter, because the downstream loss itself drives them toward near-orthogonal subspaces once they are free parameters (Section 3.4, Table 4: mean off-diagonal cosine  $\approx 0.05$  for SUM-PERCH with no penalty).

**Practical implication.** SUM-PERCH is the cheapest retrofit for legacy sum-encoding pipelines: a one-line change ( $W \rightarrow W_k$  indexed by channel),  $\mathcal{O}(C \cdot d_{\text{model}})$  extra parameters, no auxiliary loss, and none of CONCAT’s  $C \mid d_{\text{model}}$  architectural constraint. SUM-ORTHO is equivalent at convergence

Table 7: Test-time channel ablation at  $C=4$ , seed 0. We zero a single channel’s input at inference; rows show the resulting val accuracy.  $\Delta$  is accuracy drop from the unmasked baseline. The outcome formula uses ch0 in two interaction terms, ch1 in one, and ch2/ch3 once each with smaller marginal effect; block-partitioned and orthogonalised-sum encoders reproduce this ordering cleanly, whereas SUM, CI and CAT are much flatter.

| encoder      | no mask | mask ch0 | mask ch1 | mask ch2 | mask ch3 |
|--------------|---------|----------|----------|----------|----------|
| SUM          | 0.093   | 0.078    | 0.087    | 0.090    | 0.093    |
| $\Delta$ acc | —       | −0.015   | −0.006   | −0.003   | 0.000    |
| CAT          | 0.165   | 0.092    | 0.103    | 0.116    | 0.162    |
| $\Delta$ acc | —       | −0.073   | −0.062   | −0.049   | −0.003   |
| CI           | 0.140   | 0.088    | 0.121    | 0.096    | 0.141    |
| $\Delta$ acc | —       | −0.052   | −0.019   | −0.044   | +0.001   |
| FDM          | 0.148   | 0.086    | 0.099    | 0.118    | 0.146    |
| $\Delta$ acc | —       | −0.062   | −0.049   | −0.030   | −0.002   |
| CONCAT       | 0.200   | 0.091    | 0.103    | 0.119    | 0.188    |
| $\Delta$ acc | —       | −0.109   | −0.097   | −0.081   | −0.012   |
| SUM-PERCH    | 0.200   | 0.089    | 0.110    | 0.114    | 0.186    |
| $\Delta$ acc | —       | −0.111   | −0.090   | −0.086   | −0.014   |
| SUM-ORTHO    | 0.201   | 0.089    | 0.109    | 0.115    | 0.187    |
| $\Delta$ acc | —       | −0.112   | −0.092   | −0.086   | −0.014   |

but adds an auxiliary loss term that does not change the outcome on our tasks; we suggest its main use is as a mild training-speed accelerator when convergence budgets are tight.

**What the FDM carrier buys (and doesn’t).** FDM is not what it sounds like. In communications, frequency-division multiplexing prevents cross-talk on a *shared* medium; in our setting the blocks do not overlap, so there is no cross-talk to prevent in the first place. What the sinusoidal carrier actually provides is an intra-block RoPE-style positional prior: attention dot products between two positions in the same channel gain a lag-dependent cos factor without needing the model to learn the demix from the additive position channel. Our experiments show this is a *convergence-speed advantage*: FDM reaches a given NLL sooner than CONCAT early in training (faster descent), but is eventually overtaken. At convergence CONCAT wins cleanly, and at larger  $d_{\text{model}}$  the gap widens because the fixed template cannot use the extra width the way a learned projection can. A grid search over 11 fixed carrier bands and a learnable- $\omega_k$  variant both fail to close this gap; the constraint is structural, not a hyperparameter miss.

**Architectural baselines.** CI and CAT give each channel nominally more capacity than the block-partitioned encoders (full  $d_{\text{model}}$  per channel and per- $(t, k)$  token respectively) but both underperform at matched parameter count. CI shares a single backbone across channels; the bottleneck is the final-layer head’s limited capacity to combine  $C$  per-channel streams. CAT pays  $\mathcal{O}((CT)^2)$  attention and gains nothing for it on this task, presumably because the relevant structure is simultaneity of channels at a given  $t$ , not long-range temporal dependencies across channels at different times.

**What exactly fails about the sum baseline.** Our SUM baseline combines a shared scalar projection  $W \in \mathbb{R}^{d_{\text{model}}}$  with a per-channel additive embedding  $e_k$ ; this is the recipe most commonly

seen in practitioner codebases, and its total encoder capacity is  $\mathcal{O}(d_{\text{model}})$  regardless of  $C$ . Under  $15\times$  total-parameter scaling, its validation NLL moves by less than 0.02 nats. SUM-PERCH (same sum structure but with *per-channel* learned projections  $W_k$ , no orthogonality penalty) and SUM-ORTHO (per-channel  $W_k$  *plus* the penalty) let us decompose the failure: the per-channel  $W_k$  upgrade contributes most of the improvement, and the orthogonality penalty contributes the last bit. The bottleneck of SUM is therefore not summation per se—SUM-ORTHO also sums—but the combination of a shared scalar projection with a trivially small per-channel identifier (the additive embedding  $e_k$ ), which leaves no gradient signal that would encourage allocation of orthogonal subspaces.

**LayerNorm does not destroy channel separation.** Standard pre-LayerNorm could in principle mix channel blocks (it computes statistics across the full  $d_{\text{model}}$  axis), but empirically the linear probes at layer 3 recover each channel to  $R^2 \geq 0.92$ . LayerNorm rescales each coordinate by a statistic derived from all coordinates but does not linearly combine them; the directional block structure survives.

## 5 Limitations

- The benchmark is synthetic and deliberately constructed to make channel identity informative. On tasks where channels are truly exchangeable or redundant, SUM may not be penalised.
- Scales tested:  $d_{\text{model}} \in \{64, 128, 256\}$ ,  $C \in \{4, 8, 16\}$ , up to  $\sim 2.4\text{M}$  parameters. At larger model scales the interplay between encoders and overfitting may change.
- The CAT baseline is skipped at  $C=16$  for compute reasons; comparing it at matched attention budget rather than matched parameters would be a fairer test.
- Carrier hyperparameters for FDM were searched but we did not tune every design knob (carrier ladder base, per-block LayerNorm). A per-block group-norm variant may alter the FDM ceiling.
- Five seeds per configuration; for the marginal differences inside the top tier, a larger seed budget would tighten confidence intervals.

## 6 Related work

**Channel-as-token and channel-independent TSF.** iTransformer (6) transposes the problem so each variate is a token with full  $d_{\text{model}}$  capacity; PatchTST (5) runs an independent transformer per channel with shared weights; Crossformer (7) does both directions. These approaches all preserve per-channel capacity at the cost of longer sequences or per-channel compute. The block-partitioned encoding studied here sits in a different corner of the design space:  $C$  channels share one  $d_{\text{model}}$ -wide token at each time, with partitioning and identity preserved *within* the embedding rather than across tokens.

**Rotary position embedding (RoPE).** Su et al. (2) encodes positional information with rotation by frequency-graded angles. FDM re-uses the per-dimension frequency ladder but applies it within each channel’s block, with the *channel* carrier setting the outer scale.

**Perceiver IO and Fourier-feature input encodings.** Jaegle et al. (3) attaches explicit Fourier features per modality to disambiguate inputs to a shared latent attention bottleneck. FDM differs in that the modality identifier and the signal value are *multiplied* (amplitude modulation) rather than concatenated, and dimensions are physically partitioned rather than augmented.

**Orthogonality regularisation.** Soft-orthogonality penalties on weight matrices have been used for conditioning and feature decorrelation (8). SUM-ORTHO applies the same idea specifically to per-channel input projections; to our knowledge this is a new use case in the multi-signal embedding literature.

**Frequency-domain input encoders.** TimesNet (4) processes time-series in the frequency domain with an explicit fast Fourier transform (FFT). Our work is dual: we perform an inverse-FFT-like *encoding* of a single value onto a per-channel frequency, rather than transforming the signal itself.

## 7 Conclusion

On tasks where channel identity is informative, the common shared-projection summation recipe has a capacity-independent ceiling. The fix is to give each channel its own learned projection: three different dressings of this idea—free-summed (SUM-PERCH), orthogonality-regularised (SUM-ORTHO), or hard-partitioned into blocks (CONCAT)—all converge to the same downstream operating point, because the learned projections spontaneously orient into near-orthogonal subspaces under the task loss. The frequency-division analogy (FDM) provides a faster-convergence-but-lower-ceiling variant: useful as an inductive bias at small capacity, a constraint at larger ones. The architectural alternatives from the time-series forecasting literature (channel-as-token, channel-independent) underperform these three at matched parameter budget.

The most practical recommendation is SUM-PERCH: a one-line change to existing sum-encoding pipelines, no architectural constraint, no auxiliary loss. It matches everything fancier we measured.

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